WALEED IQBAL

Senior AI Engineer | Agent Evaluation, RL Environments and LLM Apps

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SUMMARY

Senior AI engineer who builds the systems that test AI agents and the LLM products that ship to real users. Core engineer on an agent evaluation platform used daily across engineering, review and delivery teams, owning its human and AI review pipeline, OSWorld desktop runs and client delivery. Built Anchorly, a live AI shopping assistant for Shopify, end to end. Strongest at reliable grading, lowering evaluation cost and securing how agents run tools. Promoted every year at Softaims from associate to senior engineer, leading delivery for three client products.

TECHNICAL SKILLS

Agent evaluation and RL: LLM evals, benchmarking, reinforcement learning environments, OSWorld V1 and V2, BrowseComp, WebArena, Harbor, Playwright, LLM as judge, rubric and state based grading, human in the loop review

LLMs and agents: OpenAI, Anthropic and Gemini APIs, Qwen3-VL and UI-TARS on vLLM, OpenRouter, tool calling, MCP servers, agent loops, RAG with pgvector, vision models

Backend: Python, FastAPI, Django, Celery, Redis, PostgreSQL, Node.js, TypeScript, BullMQ, Prisma, Ruby on Rails

Cloud, frontend and security: GCP and GKE, AWS, Docker, Kubernetes, QEMU, Prometheus, GitLab CI/CD, React, Next.js, Shopify apps, non root containers, auth hardening

PROFESSIONAL EXPERIENCE

Turing | Remote, full time contract — Sep 2023 - Present

Software Engineer, Agentic AI Evaluation Platform — *Sep 2025 - Present*

- One of the core engineers who took the agent evaluation platform to production within three months of its first commit, on FastAPI, Celery, Redis, PostgreSQL and Next.js across GCP and AWS. About 500 people use it daily, and on its busiest days it runs more than 2,000 evaluations across 10 to 15 simulated web apps.
- Built the review pipeline a client asked for: humans perform each task, an AI review scores it, and reviewers sign it off or send it back for rework, with every role and handoff permissioned and approved work shipped in the client's own format. Weekly output rose from about 100 tasks to 500 to 800, once reaching 1,000, while client rejections fell.
- Ran OSWorld desktop benchmarks on isolated Ubuntu and Windows VMs, first on GKE pods through QEMU, then on AWS after leading the move to OSWorld V2. Up to 10 desktops run at once across as many as 10 nodes, with each run taking 3 to 4 hours.
- Added VM checkpoints and state restore so graders rerun on a saved desktop instead of replaying the whole run, cutting cost per task from about $400 to $50, and let human testers work inside the same VMs with live grading.
- Integrated computer use agents from OpenAI, Anthropic and Gemini plus self hosted Qwen3-VL and UI-TARS on vLLM, mapping every provider's actions and coordinates into one trajectory format under shared rules.
- Designed layered grading that checks each result with golden references, LLM evaluators, DOM and state assertions and graders inside the environment, so results are repeatable and valid alternative solutions still pass.
- Ran 30+ long running agent tasks at once without duplicate dispatch, recovering automatically from worker loss, using atomic claims, PostgreSQL advisory locks, Redis leases and watchdog sweeps.
- Built the checks that accept a BrowseComp task only if models without internet fail it (closed book), then rerun it with search in the same chat after approval to prove its evidence holds (answer back). Integrated OpenAI and Gemini agents into Harbor for these tasks.
- Hardened the verification service after a model's MCP tools were found running as root: disabled stdio MCP execution, moved every container to a non root user, closed SSRF and authorization gaps and cleared all High and Critical dependency alerts.
- Caught a supply chain attack that hid an obfuscated payload in the ESLint config of three team repos, reported it to management and removed it from every working branch within 30 minutes.
- Reviewed teammates' pull requests and mentored new engineers through onboarding.

Software Engineer, Agentic RL Environments — *Jul 2024 - Aug 2025*

- Recreated production SaaS apps, including Salesforce, another CRM, and healthcare and travel booking apps, as offline React clones with no outside dependencies, so agent runs are reproducible.
- Designed multi step agent tasks graded on the final state of the app rather than the answer text.
- Built prompt generation for each task type and multi agent workflows that check tasks and submissions before a human reviewer sees them.

Software Engineer, LLM Code Generation and Computer Use — *Sep 2023 - Jun 2024*

- Wrote and graded Python solutions that became training and benchmark data for LLM coding models, growing from single algorithm problems to Django and React apps built over several rounds.

- Ran full development workflows with computer use agents on macOS, Ubuntu and Chromium, writing targeted prompts that exposed where models failed at reading, changing and debugging code.

Softaims | Lahore, Pakistan — Aug 2021 - Jul 2024

Senior Software Engineer — *Aug 2023 - Jul 2024*

Held concurrently with the Turing contract, with both employers' knowledge.

- Led architecture, delivery planning and releases for three concurrent client products with a team of three engineers, reporting production risk directly to clients.
- Took an upsell app rebuilt from a five year old codebase through Shopify App Store review, with theme app extensions, order history based recommendations and a GraphQL rewrite for Shopify's API transition.

Software Engineer — *Aug 2022 - Aug 2023*

- Rebuilt delivery on GitLab CI/CD with Dockerised builds to Heroku, so a non technical client running a network of German institutes could review every branch and ship it with one click.
- Gave every institute its own subdomain on a shared Rails and React codebase, rebuilt the interface on Tailwind and integrated the institutes' administration APIs.
- Built a Shopify app with Stripe billing and idempotent webhooks, so retries never charge a customer twice.

Associate Software Engineer — *Aug 2021 - Aug 2022*

- Built an embeddable WebRTC support tool that lets agents see a customer's screen with video and audio calling, plus a Chrome extension recording straight to storage via FFmpeg.
- Cut page loads on an overloaded crypto platform from about 5 seconds to under 1 by moving the UI to Next.js and realtime third party fetching onto cached Redis workers.

SELECTED PROJECTS

Anchorly: retrieval grounded AI shopping assistant for Shopify stores — 2026

Node.js, TypeScript, Shopify Remix, Prisma, PostgreSQL with pgvector, BullMQ, OpenRouter | Live: anchorlychat.com

- Syncs each store's products, pages and policies into pgvector, and describes every product photo once with a vision model so shoppers can search by how items look.
- Fetches price and stock through live tool calls, not stuffed context. Below a relevance floor it declines and takes the shopper's email. In testing on 280 products from 7 live stores, it never invented a product or price.
- Runs a one time setup on a stronger model and daily chat on low cost models, with tools scoped to what a shopper may see and built in audits that grade answers against the store's own catalog.

OPEN SOURCE PROJECTS

agentscore: evaluation harness for computer use agents — 2026

Next.js, TypeScript, Playwright, Python | Live: agentscore-sigma.vercel.app | Code: github.com/iamwaleediqbal/agentscore

- Grades the state of a live web app before and after each task, never the route taken, and keeps both snapshots so a grading change retests every past run at no model cost: 48 runs across 5 models for $0.65 in total.
- Scores agents that act on page text and on screen pixels separately, with confidence intervals: three of five models did better with text, one at 6/6 against 3/6.

polyact: one action schema across model providers — 2026

Python | Code: github.com/iamwaleediqbal/polyact

- Normalises four provider response formats into one action record in real screen pixels, and counts tokens the same way across providers, with test vectors that agentscore's runner is checked against.

clickmail: mail client built as a test environment for agents — 2026

Next.js, TypeScript | Live: clickmail-sigma.vercel.app/gym | Code: github.com/iamwaleediqbal/clickmail

- A public mail client with 52 messages in 7 folders, built for agents, with no grader inside so it can't score itself.

EDUCATION

BS Computer Science, FAST National University of Computer and Emerging Sciences, Lahore — 2017 - 2021